

L4 – 8.3 Composite Functions

MHF4U

Jensen

Two functions, f and g can be combined using a process called composition, which can be represented by:

$$f(g(x)) \text{ OR } (f \circ g)(x)$$

This is read as “ f composite g ”

Part 1: Determine the Composition of Two Functions

To determine an equation for a composite function, substitute the second function into the first.

To determine $f(g(x))$, substitute $g(x)$ in for x in to $f(x)$

Example 1: If $f(x) = x^2$ and $g(x) = x + 3$, determine an equation for each composite function and then graph the function.

a) $(f \circ g)(x)$

$$= f(g(x))$$

$$= f(x + 3)$$

$$= (x + 3)^2$$

b) $(g \circ f)(x)$

$$= g(f(x))$$

$$= g(x^2)$$

$$= x^2 + 3$$

c) $g^{-1}(g(x))$

Start by finding $g^{-1}(x)$

$$y = x + 3$$

$$x = y + 3$$

$$x - 3 = y$$

$$g^{-1}(x) = x - 3$$

Now find $g^{-1}(g(x))$

$$= g^{-1}(x + 3)$$

$$= (x + 3) - 3$$

$$= x$$

$$a) (f \circ g)(x) = (x+3)^2$$

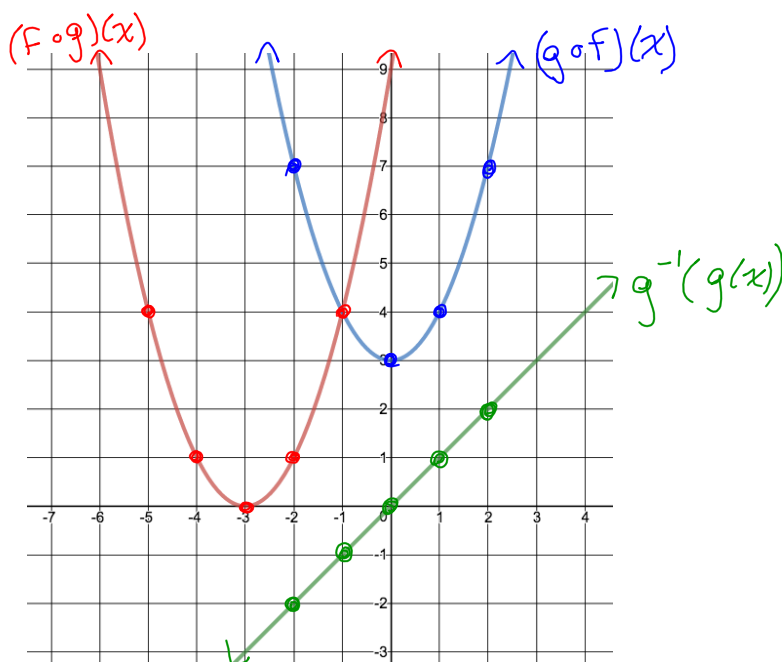
x	y
-5	4
-4	1
-3	0
-2	1
-1	4

$$b) (g \circ f)(x) = x^2 + 3$$

x	y
-2	7
-1	4
0	3
1	4
2	7

$$c) g^{-1}(g(x)) = x$$

x	y
-2	-2
-1	-1
0	0
1	1
2	2



Part 2: Evaluate a Composite Function

To evaluate a composite function $f(g(x))$ at a specific value, evaluate $g(x)$ at the specific value and then substitute the result into $f(x)$.

Example 2: If $u(x) = x^2 + 3x + 2$ and $w(x) = \frac{1}{x-1}$

a) Evaluate $(u \circ w)(2)$

b) Evaluate $w(u(-3))$

$$\begin{aligned} &= u(w(2)) \\ &= u(1) \\ &= (1)^2 + 3(1) + 2 \\ &= 6 \end{aligned}$$

$$\left\{ \begin{aligned} w(2) &= \frac{1}{2-1} \\ &= \frac{1}{1} \\ &= 1 \end{aligned} \right.$$

$$\begin{aligned} w(u(-3)) &= w(2) \\ &= \frac{1}{2-1} \\ &= 1 \end{aligned}$$

$$\left\{ \begin{aligned} u(-3) &= (-3)^2 + 3(-3) + 2 \\ &= 2 \end{aligned} \right.$$

Part 3: Application

Example 3: The number of rabbits, R , in a wildlife reserve as a function of time, t , in years can be modelled by the function $R(t) = 50 \cos(t) + 100$. The number of wolves, W , in the same reserve can be modelled by the function $W(t) = 0.2[R(t - 2)]$. Find the full equation for $W(t)$

$$R(t - 2) = 50 \cos(t - 2) + 100$$

$$W(t) = 0.2[R(t - 2)]$$

$$W(t) = 0.2[50 \cos(t - 2) + 100]$$

$$W(t) = 10 \cos(t - 2) + 20$$